Symbolic Projection Notes for $B_{c1} \to B_c^{(*)}\gamma$

Working notes

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1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \to B_c\gamma, \qquad B_{c1}(6743) \to B_c^*\gamma, \qquad B_{c1}(6750) \to B_c\gamma, \qquad B_{c1}(6750) \to B_c^*\gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s\langle\bar{s}s\rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b}\, i\gamma_5 c, \qquad j_\rho^V = \bar{b}\,\gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b}\,\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b + m_c}\,\bar{b}\,\sigma_{\mu\alpha}P^\alpha\gamma_5 c, \qquad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \qquad J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B.$$

This mixing is used at the invariant-amplitude level. The OPE is first derived for the basis currents because the two Dirac traces are independent; the physical amplitudes are then

$$\widehat{\Pi}_1 = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_2 = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the pseudoscalar final state,

$$g_i = \frac{m_b + m_c}{m_{B_c}^2 f_{B_c}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c}^2}{M_2^2}\right)\widehat{\Pi}_i^{(P)},$$

while for the vector final state,

$$g_i^* = \frac{1}{m_{B_c^*} f_{B_c^*}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c^*}^2}{M_2^2}\right)\widehat{\Pi}_i^{(V)}.$$

The pilot numerical script applies this rotation. The temporary approximation is instead in the normalization: a common benchmark $f_{B_{c1}} = 0.373$ GeV is used for both physical axial-vector residues, pending a separate f_1, f_2 two-point extraction.

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \qquad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q)\, ie_Q\gamma_\nu\epsilon^\nu\, S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}} = +e/3$, while $e_c = +2e/3$. Equivalently, one may keep $e_b = -e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c-mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted B_c-mixing work uses the same local basis,

$$J_\mu^A = \bar{b}\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b + m_c}\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5 c,$$

with $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, and the final working window

$$7\ \text{GeV}^2 \le M^2 \le 9\ \text{GeV}^2, \qquad 53\ \text{GeV}^2 \le s_0 \le 55\ \text{GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

3 Step-by-Step Sum-Rule Construction

This section is written as a checklist for reproducing the calculation.

1. Start from basis currents. The physical currents J_1, J_2 are mixed combinations of J_A, J_B. We derive the OPE for J_A and J_B separately only because the Dirac traces are simpler. The physical amplitudes are obtained afterward through the $\sin\theta, \cos\theta$ rotation.

2. Write the correlators. For $B_c\gamma$,

$$\Pi_\mu^{(5,K)}(p,q) = i\int d^4x\, e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)i\gamma_5 c(x)\, J_{K\mu}^\dagger(0)\}|0\rangle, \qquad K = A, B.$$

For $B_c^*\gamma$,

$$\Pi_{\mu\nu}^{(V,K)}(p,q) = i\int d^4x\, e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)\gamma_\nu c(x)\, J_{K\mu}^\dagger(0)\}|0\rangle.$$

3. Contract the heavy fields. There are two hard-photon diagrams: photon emission from the c line and photon emission from the $\bar{b}$ line. The free heavy propagator is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0},$$

and the photon insertion is

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q)\, ie_Q\not{\epsilon}\, S_Q^{(0)}(k).$$

The charge convention in the scripts is $e_c = +2/3$ and $e_{\bar{b}} = +1/3$. No photon DA or light-quark condensate term appears in this heavy-heavy leading baseline.

4. Evaluate and project the traces. For $B_c\gamma$ project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p\cdot q)^2}, \qquad S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

For $B_c^*\gamma$ the current diagnostic tensor is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho p q} - (p\cdot q)\epsilon_{\mu\nu\rho p}, \qquad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p\cdot q)^2.$$

The vector-channel result still needs matching to the physical $1^+ \to 1^-\gamma$ amplitude basis.

5. Reduce to denominator structures. For photon emission from the charm line use

$$D_1 = (k+q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k-p)^2 - m_b^2.$$

For photon emission from the anti-bottom line use

$$D_1 = k^2 - m_c^2, \quad D_2 = (k-p)^2 - m_b^2, \quad D_3 = (k-p+q)^2 - m_b^2.$$

The triangle pieces contain the full three-denominator product. Terms where a numerator cancels one denominator become two-denominator/contact terms and must be included or shown to vanish.

6. Borel match and rotate. The true variables are p^2 and $P^2 = (p+q)^2$, with

$$p\cdot q = \frac{P^2 - p^2}{2}.$$

The pilot numerical calculation uses the diagonal choice $M_1^2 = M_2^2 = 2M^2$, leading to the practical exponential

$$\exp\left[\frac{m_i^2 + m_f^2}{2M^2}\right].$$

After obtaining $\widehat{\Pi}_A, \widehat{\Pi}_B$, the physical combinations are

$$\widehat{\Pi}_{6743} = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_{6750} = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the $B_c\gamma$ widths we use

$$\Gamma(1^+ \to 0^-\gamma) = \frac{\alpha_{\rm em}}{3} g^2 E_\gamma^3, \qquad E_\gamma = \frac{m_i^2 - m_f^2}{2m_i}.$$

4 $B_{c1} \to B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p\cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \qquad p\cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$\begin{aligned} C_A^{(c)} &= -4i\,m_c + \frac{4i\,m_b m_c^2}{pq} - \frac{4i\,m_c^3}{pq}, \\ C_A^{(\bar b)} &= -4i\,m_b + \frac{4i\,m_b^3}{pq} - \frac{4i\,m_b^2 m_c}{pq}, \end{aligned}$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar b} C_A^{(\bar b)}.$$

For the tensor current,

$$\begin{aligned} C_B^{(c)} &= \frac{4i\,m_b m_c - 8i\,m_c^2}{m_b + m_c} - \frac{4i\,m_c^2 p_2}{(m_b+m_c)pq}, \\ C_B^{(\bar b)} &= -\frac{4i\,m_b m_c}{m_b+m_c} - \frac{4i\,m_b^2 p_2}{(m_b+m_c)pq}, \end{aligned}$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar b} C_B^{(\bar b)}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

5 $B_{c1} \to B_c^* \gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu\, \epsilon_{\mu\rho pq} - (p\cdot q)\, \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p_2 (pq)^2, \qquad V_{\mu\nu\rho}\mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$\begin{aligned} C_{A,V}^{(c)} &= -\frac{4i\,m_b m_c}{p_2} + \frac{2i\,m_c^2}{p_2} + \frac{4i\,m_c^2}{pq}, \\ C_{A,V}^{(\bar b)} &= -\frac{2i\,m_b^2}{p_2} + \frac{4i\,m_b m_c}{p_2} + \frac{4i\,m_b^2}{pq}. \end{aligned}$$

The tensor-current vector cores are

$$\begin{aligned} C_{B,V}^{(c)} &= \frac{2i\,m_c}{m_b+m_c} + \frac{2i\,m_b^2 m_c - 4i\,m_b m_c^2 + 2i\,m_c^3}{(m_b+m_c)p_2} \\ &\quad - \frac{4i\,m_b m_c^2 - 4i\,m_c^3}{(m_b+m_c)pq} + \frac{2i\,m_c\,pq}{(m_b+m_c)p_2}, \\ C_{B,V}^{(\bar b)} &= \frac{2i\,m_b}{m_b+m_c} + \frac{-6i\,m_b^3 + 4i\,m_b^2 m_c + 2i\,m_b m_c^2}{(m_b+m_c)p_2} \\ &\quad - \frac{4i\,m_b^3 - 4i\,m_b^2 m_c}{(m_b+m_c)pq} + \frac{2i\,m_b\,pq}{(m_b+m_c)p_2}. \end{aligned}$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar b} C_{A,V}^{(\bar b)}, \qquad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar b} C_{B,V}^{(\bar b)}.$$

6 First Numerical Hard-Photon Pass

The script `Bc_gamma/scripts/bc_radiative_pilot_sumrule.py` uses the compact triangle cores above to produce a first hard-photon numerical estimate. For this pass we use the same central heavy-quark inputs and Borel window as the submitted B_c-mixing analysis,

$$m_b = 4.18 \text{ GeV}, \qquad m_c = 1.27 \text{ GeV}, \qquad \theta = 43.3^\circ,$$

more precisely $\theta_{\text{mean}} = 43.295^\circ$ with $\sigma_\theta = 0.151^\circ$ and 16–84% interval $[43.147^\circ, 43.455^\circ]$ from the B_c-mixing Monte Carlo. The pilot table uses the rounded central value 43.3°; the θ uncertainty is not yet included in the quoted radiative intervals.

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \qquad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The decay-constant inputs are

$$f_{B_c} = 0.371 \text{ GeV}, \qquad f_{B_c^*} = 0.384 \text{ GeV}, \qquad f_{B_{c1}} = 0.373 \text{ GeV}.$$

The first two are standard external benchmark inputs. The last is a common axial-vector benchmark used only for this first pass; separate physical f_1 and f_2 residues have not yet been extracted from a two-point mixed-current B_c sum rule in this folder.

Table 1: Pilot hard-photon results in the B_c sector. The intervals show the envelope over $M^2 = \{7, 8, 9\}$ GeV^2 and $s_0 = \{53, 54, 55\}$ GeV^2.

Channel	Status	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \to B_c\gamma$	hard-QCDSR baseline	$+0.0219\,[0.0209, 0.0232]$ GeV^{-1}	$0.108\,[0.098, 0.121]$
$B_{c1}(6750) \to B_c\gamma$	hard-QCDSR baseline	$-0.354\,[-0.387, -0.331]$ GeV^{-1}	$29.6\,[25.9, 35.3]$
$B_{c1}(6743) \to B_c^*\gamma$	diagonal vector pilot	$-0.0620\,[-0.0669, -0.0583]$	$46.3\,[41.0, 53.8]$
$B_{c1}(6750) \to B_c^*\gamma$	diagonal vector pilot	$+0.362\,[0.335, 0.401]$	$1.67\,[1.43, 2.04] \times 10^3$

The $B_c\gamma$ entries are the controlled hard-photon baseline from the present calculation. The $B_c^*\gamma$ entries should be read as diagnostic numbers only. They are obtained from the gauge-invariant $V_{\mu\nu\rho}$ projection and a diagonal single-variable reduction. Before using them as final paper results one must match this coefficient to the physical $1^+ \to 1^-\gamma$ tensor basis and carry out the full double-Borel analysis, including denominator-cancellation or contact terms.

7 Comparison with Experiment and Other Theory

LHCb has observed a wide peaking structure in the $B_c^+\gamma$ mass spectrum with significance above seven standard deviations. A minimal two-peak description gives [1, 2]

$$M_1 = 6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}, \qquad M_2 = 6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV},$$

where the errors are statistical, systematic, and due to the B_c^+ mass. No individual radiative partial width is measured. The natural widths are expected to be only a few hundred keV and are neglected in the experimental fit because they are small compared with the photon-energy resolution.

The lower visible peak should not be identified one-to-one with the input 6743 MeV state without care. In the measured $B_c\gamma$ spectrum, decays through $B_c^*\gamma$, followed by an unreconstructed soft $B_c^* \to B_c\gamma$, can shift visible peaks by the unknown hyperfine splitting.

For comparison with other theory, Martín-González et al. [3] quote representative quark-model E1 widths for the row labelled $B_{c1}(1P)$:

$$\Gamma[B_{c1}(1P) \to B_c\gamma] = 1.5 \times 10^{-4} \text{ keV}, \qquad \Gamma[B_{c1}(1P) \to B_c^*\gamma] = 146 \text{ keV}.$$

Table 2: Qualitative comparison with the current experimental information from LHCb [1, 2].

Quantity	Experiment	Present input/result
Observed $B_c(1P)^+$-like signal	yes, $> 7\sigma$ in $B_c^+\gamma$	channels studied explicitly
Lower visible peak	$6704.8 \pm 5.5 \pm 2.8 \pm 0.3$ MeV	input state 6743 MeV
Higher visible peak	$6752.4 \pm 9.5 \pm 3.1 \pm 0.3$ MeV	input state 6750 MeV
Partial widths	not measured	$B_c\gamma$: 0.108, 29.6 keV

Their table also lists earlier model values from Refs. [4, 5, 6],

$$\Gamma[B_{c1}(1P) \to B_c\gamma] = 13,\ 18.4,\ 0.0 \text{ keV},$$

$$\Gamma[B_{c1}(1P) \to B_c^*\gamma] = 60,\ 78.9,\ 99.5 \text{ keV}.$$

The comparison therefore reads as follows.

Table 3: Comparison with representative E1 model widths. The model range is collected from Ref. [3], which includes comparisons with Refs. [4, 5, 6].

Channel	Other-theory scale	This work	Interpretation
$B_{c1} \to B_c\gamma$	0 to 18.4 keV	0.108 keV for 6743, 29.6 keV for 6750	same broad scale, but state assignment and mixing differ
$B_{c1} \to B_c^*\gamma$	60 to 146 keV	46.3 keV for 6743, 1.67 MeV for 6750	lower pilot plausible; upper pilot flags unfinished vector normalization

Thus the $B_c\gamma$ baseline is already useful for comparison, while the $B_c^*\gamma$ channel must still be completed with the physical tensor-basis normalization and the full double-Borel/contact-term treatment.

8 Next Step

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \qquad pq = \frac{P^2 - p^2}{2}.$$

For $B_c\gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^*\gamma$, the additional $1/p_2$ and pq/p_2 structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.

Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k+q)^2 - m_c^2, \qquad D_2 = k^2 - m_c^2, \qquad D_3 = (k-p)^2 - m_b^2.$$

With Feynman parameters x, y, z, $x + y + z = 1$, the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1-z)m_c^2 + zm_b^2 - zy\,p^2 - xz\,P^2.$$

For photon emission from the $\bar{b}$ line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \qquad D_2 = (k-p)^2 - m_b^2, \qquad D_3 = (k-p+q)^2 - m_b^2,$$

and the analogous reduction gives, with $x + y + z = 1$,

$$\Delta_{\bar{b}} = xm_c^2 + (1-x)m_b^2 - x(1-x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in p^2 and P^2, not simply the two-point mixing correlator with an extra charge factor.

References

[1] R. Aaij et al. (LHCb Collaboration), "Observation of Orbitally Excited B_c^+ States," Phys. Rev. Lett. 135, 231902 (2025).

[2] R. Aaij et al. (LHCb Collaboration), "Study of $B_c(1P)^+$ states in the $B_c^+\gamma$ mass spectrum," Phys. Rev. D 112, 112003 (2025).

[3] B. Martín-González, P. G. Ortega, D. R. Entem, F. Fernández, and J. Segovia, "Towards the discovery of novel B_c states: radiative and hadronic transitions," arXiv:2205.05950.

[4] S. Godfrey, Phys. Rev. D 70, 054017 (2004).

[5] D. Ebert, R. N. Faustov, and V. O. Galkin, Phys. Rev. D 67, 014027 (2003).

[6] E. J. Eichten and C. Quigg, Phys. Rev. D 49, 5845 (1994).